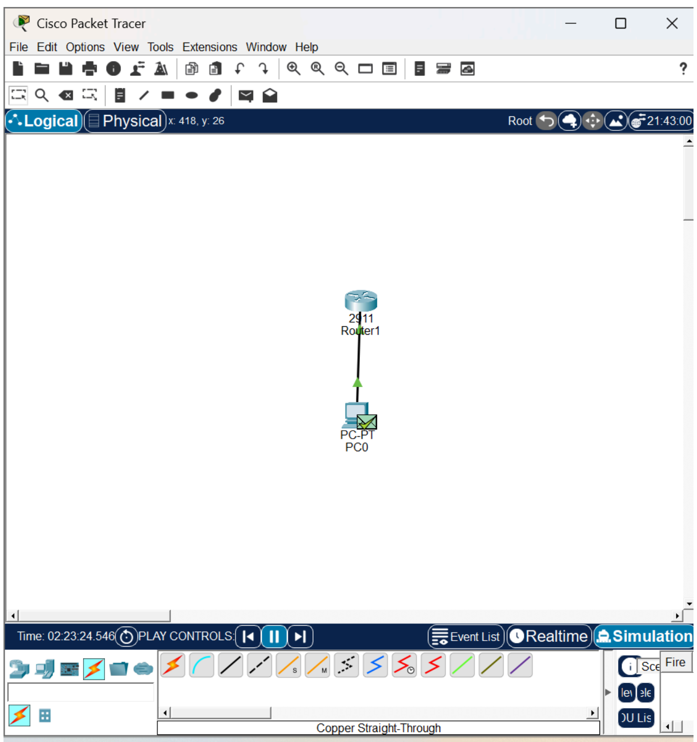
SSH Remote Access Lab

Objective

The purpose of this lab was to configure secure remote access to a router using SSH instead of Telnet. SSH ensures that communication between devices is encrypted.

Network Topology

The network consists of one router (R1) and one PC (PC-A) connected on the same network.

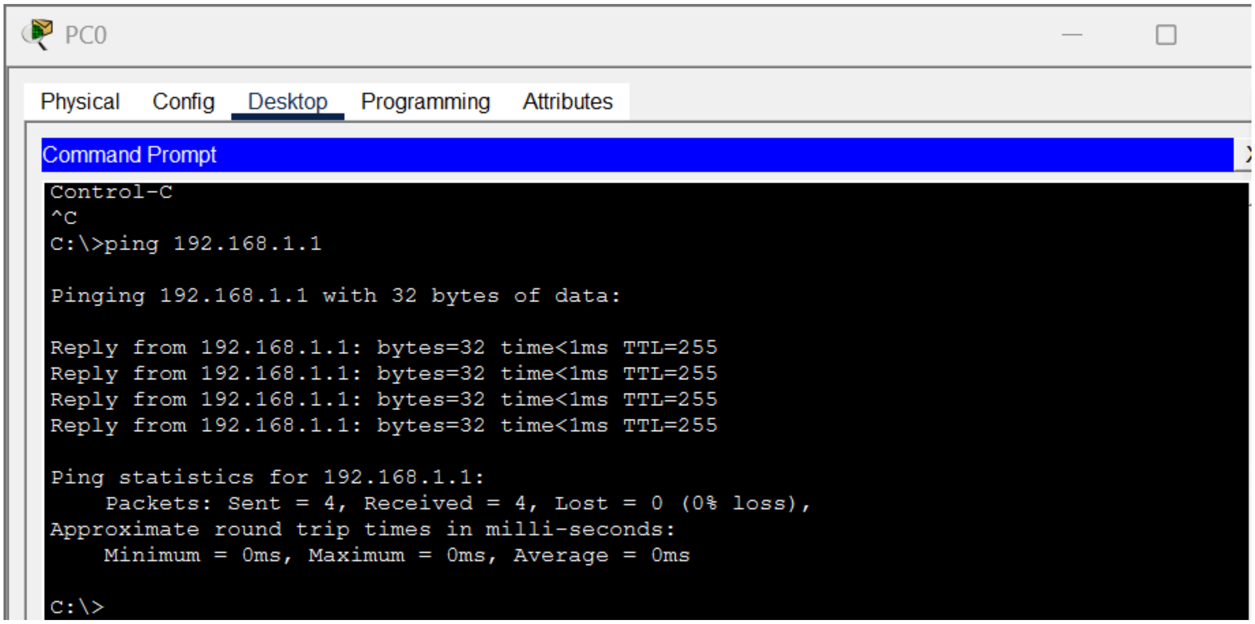


IP Address Configuration

The following IP addresses were assigned:

- R1: 192.168.1.1
- PC-A: 192.168.1.10

A ping test from the PC to the router confirmed connectivity.



SSH Configuration

SSH was configured on the router by setting a hostname and domain name, creating a local user account, generating RSA keys, enabling SSH version 2, and restricting remote access to SSH only.

SSH Connection

An SSH session was initiated from the PC to the router using the configured credentials. The connection was successful.

```
C:\>ssh -l admin 192.168.1.1

Password:
% Password:  timeout expired!
% Login invalid

[Connection to 192.168.1.1 closed by foreign host]
C:\>ssh -l admin 192.168.1.1

Password:
% Login invalid


Password:



R1>username admin secret cisco123
            ^
% Invalid input detected at '^' marker.

R1>
```

Verification

The following commands were used to verify the configuration:

- show ip ssh
- show ssh

These confirmed that SSH was enabled and that an active session existed.

```
R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1#
```

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```
R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1#show ssh
Connection      Version Mode Encryption       Hmac      State              Username
0               1.99    IN   aes128-cbc     hmac-sha1     Session Started
admin
0               1.99    OUT  aes128-cbc     hmac-sha1     Session Started
admin
%No SSHv1 server connections running.
R1#
```

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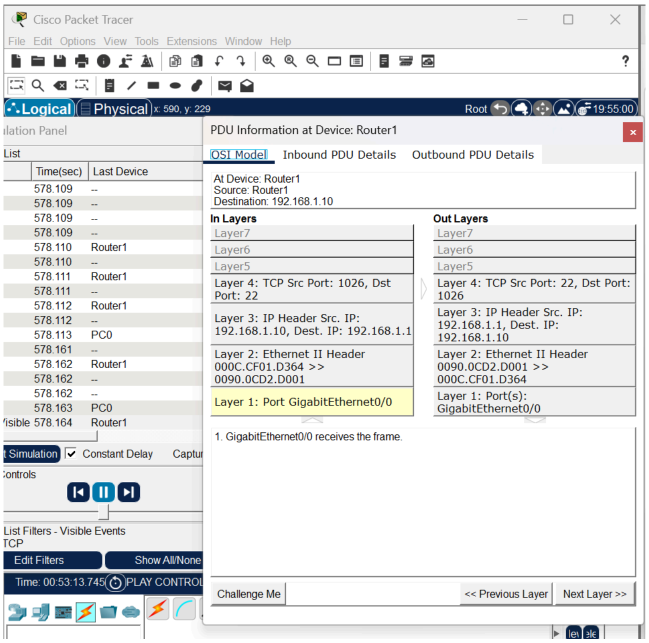
Simulation Mode

Simulation Mode in Packet Tracer was used to capture traffic during the SSH session. The packets showed SSH over TCP using port 22, confirming encrypted communication.



Conclusion

SSH was successfully configured and verified. Secure remote access was established, and encrypted traffic was confirmed.